

Numerical methods

Lecture 2: Error analysis, Solving systems of linear equations,
Gaussian elimination

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ELTE IK

- ① Elements of error analysis
- ② Applications of systems of linear equations
- ③ Systems of linear equations
- ④ Gaussian elimination

- 1 Elements of error analysis
- 2 Applications of systems of linear equations
- 3 Systems of linear equations
- 4 Gaussian elimination

Definition: Error measurements

Let A be an accurate value, and a its approximated value. In this case:

$\Delta a := A - a$ is the (exact) error,

$|\Delta a| := |A - a|$ is the absolute error of the approximation,

$\Delta_a \geq |\Delta a|$ is an error bound of a ,

$\delta a := \frac{\Delta a}{A} \approx \frac{\Delta a}{a}$ is the relative error of a ,

$\delta_a \geq |\delta a|$ is a relative error bound of a .

Example

Apply these notions in case of the approximating value 3.14 of π (rounded to two digits)!

Example

What is the value of $\sqrt{2021} - \sqrt{2020}$?

(Having the square root values rounded to 8 digits.)

May be calculated as:

$$\begin{aligned}\sqrt{2021} - \sqrt{2020} &= (\sqrt{2021} - \sqrt{2020}) \cdot \frac{\sqrt{2021} + \sqrt{2020}}{\sqrt{2021} + \sqrt{2020}} = \\ &= \frac{2021 - 2020}{\sqrt{2021} + \sqrt{2020}} = \frac{1}{\sqrt{2021} + \sqrt{2020}}.\end{aligned}$$

Let us try both ways!

0.011123480000002

0.011123477488896

Theorem: Error bounds for the basic arithmetic operations

$$\Delta_{a \pm b} = \Delta_a + \Delta_b$$

$$\delta_{a \pm b} = \frac{|a| \cdot \delta_a + |b| \cdot \delta_b}{|a \pm b|}$$

$$\Delta_{a \cdot b} = |b| \cdot \Delta_a + |a| \cdot \Delta_b$$

$$\delta_{a \cdot b} = \delta_a + \delta_b$$

$$\Delta_{a/b} = \frac{|b| \cdot \Delta_a + |a| \cdot \Delta_b}{b^2}$$

$$\delta_{a/b} = \delta_a + \delta_b$$

Remark: the resulting error bounds may become significantly (orders of magnitude) larger than the error bounds of the initial values in two cases:

- 1 δ_{a-b} : when subtracting nearby values.
- 2 $\Delta_{a/b}$: when dividing by a small number.

Avoid these kinds of operations in implementations at all cost!
(Or suffer from unacceptably large errors. . .)

Proof. (Consider now addition and subtraction of values with the same sign.)

The error of $a \pm b$:

$$\Delta(a \pm b) = (A \pm B) - (a \pm b) = (A - a) \pm (B - b) = \Delta a \pm \Delta b$$

$$|\Delta(a \pm b)| = |\Delta a \pm \Delta b| \leq |\Delta a| + |\Delta b| \leq \Delta_a + \Delta_b = \Delta_{a \pm b}.$$

The relative error of $a \pm b$:

$$\frac{\Delta(a \pm b)}{a \pm b} = \frac{\Delta a \pm \Delta b}{a \pm b} = \frac{a \cdot \delta a \pm b \cdot \delta b}{a \pm b}$$

$$\begin{aligned} \frac{|\Delta(a \pm b)|}{|a \pm b|} &= \frac{|a \cdot \delta a \pm b \cdot \delta b|}{|a \pm b|} \leq \frac{|a| \cdot |\delta a| + |b| \cdot |\delta b|}{|a \pm b|} \leq \\ &\leq \frac{|a| \cdot \delta_a + |b| \cdot \delta_b}{|a \pm b|} = \delta_{a \pm b} \end{aligned}$$

Error propagation: multiplication

The error of multiplication:

$$\begin{aligned}\Delta(a \cdot b) &= A \cdot B - a \cdot b = A \cdot B - A \cdot b + A \cdot b - a \cdot b = \\ &= A(B - b) + b(A - a) = A \cdot \Delta b + b \cdot \Delta a = \\ &= (a + \Delta a) \cdot \Delta b + b \cdot \Delta a \approx a \cdot \Delta b + b \cdot \Delta a \\ &\quad (\Delta a \cdot \Delta b \text{ is neglected})\end{aligned}$$

$$|\Delta(a \cdot b)| \leq |a| \cdot |\Delta b| + |b| \cdot |\Delta a| \leq |a| \cdot \Delta_b + |b| \cdot \Delta_a = \Delta_{a \cdot b}$$

As for the relative error:

$$\delta(a \cdot b) = \frac{\Delta(a \cdot b)}{a \cdot b} \approx \frac{a \cdot \Delta b + b \cdot \Delta a}{a \cdot b} = \frac{\Delta b}{b} + \frac{\Delta a}{a} = \delta b + \delta a$$

$$|\delta(a \cdot b)| \leq |\delta a| + |\delta b| \leq \delta_a + \delta_b = \delta_{a \cdot b}$$

The error of division:

$$\begin{aligned}
 \Delta \left(\frac{a}{b} \right) &= \frac{A}{B} - \frac{a}{b} = \frac{A \cdot b - a \cdot B}{Bb} = \\
 &= \frac{A \cdot b - a \cdot b + a \cdot b - a \cdot B}{Bb} = \frac{b \cdot (A - a) - a \cdot (B - b)}{Bb} = \\
 &= \frac{b \cdot \Delta a - a \cdot \Delta b}{(b + \Delta b) \cdot b} \approx \frac{b \cdot \Delta a - a \cdot \Delta b}{b^2} \\
 &(\Delta b \cdot b \text{ is neglected})
 \end{aligned}$$

$$\left| \Delta \left(\frac{a}{b} \right) \right| \leq \frac{|b| \cdot |\Delta a| + |a| \cdot |\Delta b|}{b^2} \leq \frac{|b| \cdot \Delta_a + |a| \cdot \Delta_b}{b^2} = \Delta_{a/b}$$

The relative error of division:

$$\begin{aligned}\delta\left(\frac{a}{b}\right) &= \frac{\Delta\left(\frac{a}{b}\right)}{\frac{a}{b}} \approx \frac{b \cdot \Delta a - a \cdot \Delta b}{b^2} \cdot \frac{b}{a} = \\ &= \frac{b \cdot \Delta a - a \cdot \Delta b}{b \cdot a} = \frac{\Delta a}{a} - \frac{\Delta b}{b} = \\ &= \delta a - \delta b = \delta\left(\frac{a}{b}\right)\end{aligned}$$

$$|\delta\left(\frac{a}{b}\right)| \leq |\delta a| + |\delta b| \leq \delta_a + \delta_b = \delta_{a/b}$$



Remark:

- Use error bounds **as sharp** (as small) **as possible!**
- Use **easy to handle** error bounds! (E.g. 0.187 vs. 0.2)

Example 1

What is the value of $\exp(\pi)$?

Using the approximated value of π rounded to 2 and 15 digits:

23.140692632779267

23.103866858722185

Example 2

What is the value of $\log(\pi)$?

Using the approximated value of π rounded to 2 and 15 digits:

1.144729885849400

1.144222799920162

Which function is more 'tolerant' about the error of the argument?

Theorem: Error of functions I.

If $f \in C^1(b_{\Delta_a}(a))$ with $b_{\Delta_a}(a) = [a - \Delta_a; a + \Delta_a]$, then

$$\Delta_{f(a)} = M_1 \cdot \Delta_a,$$

with $M_1 = \max \{ |f'(\xi)| : \xi \in b_{\Delta_a}(a) \}$.

Proof. Use the Lagrange mean value theorem.

$$\Delta f(a) = f(A) - f(a) = f'(\xi) \cdot (A - a) = f'(\xi) \cdot \Delta a,$$

for some value $\xi \in b_{\Delta_a}(a)$. For the absolute error (bound):

$$|\Delta f(a)| = |f'(\xi)| \cdot |\Delta a| \leq M_1 \cdot \Delta_a = \Delta_{f(a)}. \quad \square$$

Theorem: Error of functions 2.

If $f \in C^2(b_{\Delta_a}(a))$ with $b_{\Delta_a}(a) = [a - \Delta_a; a + \Delta_a]$, then

$$\Delta_{f(a)} = |f'(a)| \Delta_a + \frac{M_2}{2} \cdot \Delta_a^2,$$

with $M_2 = \max \{ |f''(\xi)| : \xi \in b_{\Delta_a}(a) \}$.

Proof. Use the Taylor formula.

$$\Delta f(a) = f(A) - f(a) = f'(a) \cdot (A - a) + \frac{f''(\xi)}{2} \cdot (A - a)^2,$$

for some value $\xi \in b_{\Delta_a}(a)$.

For the absolute error (bound):

$$\begin{aligned} |\Delta f(a)| &= |f'(a)| \cdot |\Delta a| + \frac{|f''(\xi)|}{2} \cdot |\Delta a|^2 \leq \\ &\leq |f'(a)| \cdot \Delta_a + \frac{M_2}{2} \cdot \Delta_a^2 = \Delta_{f(a)}, \end{aligned}$$



Corollary: Relative error of a function value

If Δ_a is small, then $\delta_{f(a)} = \frac{|a| |f'(a)|}{|f(a)|} \cdot \delta_a$.

Proof. If Δ_a is small, then Δ_a^2 may be neglected (in Theorem 2 above), thus, using $\Delta_a = |a| \cdot \delta_a$

$$|\delta f(a)| \approx \frac{|f'(a)| \cdot \Delta_a}{|f(a)|} = \frac{|a| \delta_a \cdot |f'(a)|}{|f(a)|} = \frac{|a| |f'(a)|}{|f(a)|} \cdot \delta_a.$$

□

Definition: Condition number of function f at point a

The value $c(f, a) = \frac{|a| |f'(a)|}{|f(a)|}$ is called the condition number of function f at point a .

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- **In elementary school:**

Mathematics competition problem (3rd grade)

Each letter in the word MATHS denotes a one digit number.

For these numbers we have:

$$M + A + T + H + S = 25$$

$$M + A = 11$$

$$A + T = 10$$

$$T + H = 12$$

$$H + S = 10$$

Which letter denotes which number, assuming that the 5 letters denote different digits?

- **Economical calculations:**

Imagine a factory building output products P_1 and P_2 from input products P_5 and P_6 , maybe with intermediate products A_3, A_4 . The **input-output matrix** A may be formed. . . (see economics)

The other way around? (How much of this input is present in one final product?) The answer is given by the **Leontief inverse** $T = (I - A)^{-1}$.

With these matrices:

From inputs x we get $y = (I - A) \cdot x$ output products, and for y output we need $x = T \cdot y = (I - K)^{-1} \cdot x$ input products.

- Numerical solutions to engineering problems
(c.f. the oscillating string)
- Interpolation with spline functions
(sound equalizer, image resizing, easings)
- Approximation problems

- **Stationary models of networks:** electric networks, circuits, water and gas networks etc. modeled as directed graphs. Numbers on the edges represent flow intensity (of electrons, water, gas etc.)
Sources and sinks may exist.
Kirchhoff's law in each vertex: flow in and flow out cancels.
A linear system of equations describes the network's flow intensity values.

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Systems of linear equations: reminder

System of linear equations, Linear system (of equations)

Short: SLE, LSE, LS

Regular (scalar) form:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \cdots & + & a_{nn}x_n & = & b_n \end{array}$$

n equations, n unknowns

Remark. The number of unknowns is the same as the number of equations!

Systems of linear equations: reminder

Matrix form:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix},$$

i.e.

$$Ax = b \quad A \in \mathbb{R}^{n \times n}, \quad b, x \in \mathbb{R}^n.$$

Task:

A and b is given, find x .

Systems of linear equations: reminder

Theorem: Reminder from Basic Mathematics

- A SLE has a solution $\iff b$ can be written as a linear combination of the column vectors of A .
- A unique solution exist $\iff$ the columns of A are linearly independent $\iff \text{rank}(A) = n \iff \det(A) \neq 0 \iff A$ is invertible ($x = A^{-1}b$).

Remark:

- If A has a special form (e.g. diagonal or triangular), then the system is easy to solve. (This is one of the basic ideas.)
- Cramer's rule may be applied for at most 3×3 matrices.

- Direct methods, decompositions (finite steps, 'exact' solution)
 - Gaussian elimination, Progonka method (tridiagonal case)
 - LU decomposition, LDU , LL^T , Cholesky
 - QR decomposition (Gram–Schmidt, Householder)
 - ILU decomposition
- Iteration methods (a converging sequence of vectors)
 - matrix norms, Banach fixed point theorem
 - Jacobi iteration, Gauss–Seidel iteration, Richardson iteration
 - ILU algorithm
- Variation methods (minimizing an objective function)
 - gradient method
 - conjugate gradient method

Direct methods

- Gaussian elimination,
- Progonka method (quick Gaussian elimination for tridiagonal systems),
- LU decomposition.

Properties of direct methods:

- Exact solution in case of exact calculations.
- Finite operation count.
- Approximate calculations (e.g. finite arithmetics) may lead to non-resolvable errors in the result.

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Example: Solving a SLE using GE

Solve the following system of linear equations using Gaussian elimination!

$$\begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} \cdot x = \begin{bmatrix} -1 \\ 3 \\ -3 \end{bmatrix}$$

Gaussian elimination: an example

Elimination (left-to-right):

(Usually we write a vertical line before the right-hand-side vector.)

Step 1:

$$\text{Row 2} := \text{Row 2} - \underbrace{\left(\frac{-4}{2}\right)}_{+2} \cdot \text{Row 1}$$

$$\text{Row 3} := \text{Row 3} - \underbrace{\left(\frac{6}{2}\right)}_{-3} \cdot \text{Row 1}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 3 & -1 \\ -4 & 5 & -2 & 3 \\ 6 & -5 & 4 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 2 & 0 & 3 & -1 \\ 0 & 5 & 4 & 1 \\ 0 & -5 & -5 & 0 \end{array} \right] \rightsquigarrow$$

Gaussian elimination: an example

Step 2:

$$\text{Row 3} := \text{Row 3} - \underbrace{\left(\frac{-5}{5}\right)}_{+1} \cdot \text{Row 2}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 3 & -1 \\ 0 & 5 & 4 & 1 \\ 0 & -5 & -5 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 2 & 0 & 3 & -1 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right] \rightsquigarrow$$

Back substitution:

Row 3 := Row 3 / (-1)

Row 2 := Row 2 - 4 · new Row 3

Row 1 := Row 1 - 3 · new Row 3

$$\left[\begin{array}{ccc|c} 2 & 0 & 3 & -1 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 2 & 0 & 0 & 2 \\ 0 & 5 & 0 & 5 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

Gaussian elimination: an example

Row 2 := Row 2 / 5

Row 1 := Row 1 - 0 · Row 2

Row 1 := Row 1 / 2.

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 2 \\ 0 & 5 & 0 & 5 \\ 0 & 0 & 1 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

So the solution of the system is the vector $\mathbf{x} = [1, 1, -1]^T$. (Verify!)

Gaussian elimination: preparations

Let $a_{i,n+1} := b_i$, i.e. store the system as $[A|b]$.

GE := Gaussian elimination

$$A^{(0)} := \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & a_{1,n+1} = b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & a_{2,n+1} = b_2 \\ \vdots & & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & a_{n,n+1} = b_n \end{array} \right]$$

Goal: Rewrite the SLE into a more simple form:

- 1 left-to-right: cancel the elements below the diagonal,
'forward', GE
- 2 right-to-left: cancel above the diagonal,
'reverse', back substitution

Gaussian elimination: Step 1

Leave equation 1 unchanged.

If $a_{11}^{(0)} \neq 0$, then in order to cancel $a_{i1}^{(0)}$, subtract $\left(\frac{a_{i1}^{(0)}}{a_{11}^{(0)}}\right)$ times equation 1 from equation i ($i = 2, 3, \dots, n$).

($\rightsquigarrow$ elimination, cancellation)

$$A^{(1)} = \left[\begin{array}{ccccc|c} a_{11}^{(0)} & a_{12}^{(0)} & a_{13}^{(0)} & \cdots & a_{1n}^{(0)} & a_{1,n+1}^{(0)} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2n}^{(1)} & a_{2,n+1}^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & a_{n3}^{(1)} & \cdots & a_{nn}^{(1)} & a_{n,n+1}^{(1)} \end{array} \right],$$

where

$$a_{ij}^{(1)} = a_{ij}^{(0)} - \frac{a_{i1}^{(0)}}{a_{11}^{(0)}} \cdot a_{1j}^{(0)} \quad (i = 2, \dots, n; j = 2, \dots, n, n+1).$$

Gaussian elimination: Step 2

Leave equations 1 and 2 unchanged.

If $a_{22}^{(1)} \neq 0$, then in order to cancel $a_{i2}^{(1)}$, subtract $\left(\frac{a_{i2}^{(1)}}{a_{22}^{(1)}}\right)$ times equation 2 from equation i ($i = 3, 4, \dots, n$).

$$A^{(2)} = \left[\begin{array}{ccccc|c} a_{11}^{(0)} & a_{12}^{(0)} & a_{13}^{(0)} & \cdots & a_{1n}^{(0)} & a_{1,n+1}^{(0)} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2n}^{(1)} & a_{2,n+1}^{(1)} \\ 0 & 0 & a_{33}^{(2)} & \cdots & a_{3n}^{(2)} & a_{3,n+1}^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & a_{n3}^{(2)} & \cdots & a_{nn}^{(2)} & a_{n,n+1}^{(2)} \end{array} \right],$$

where

$$a_{ij}^{(2)} = a_{ij}^{(1)} - \frac{a_{i2}^{(1)}}{a_{22}^{(1)}} \cdot a_{2j}^{(1)} \quad (i = 3, \dots, n; j = 3, \dots, n, n+1).$$

Gaussian elimination: Step k

Leave equations $1, 2, \dots, k$ unchanged.

If $a_{kk}^{(k-1)} \neq 0$, then in order to cancel $a_{ik}^{(k-1)}$, subtract $\left(\frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \right)$ times equation k from equation i ($i = k + 1, \dots, n$). We saw this when writing Step 2, using the results after Step 1. (Write k instead of 2 to get the general formula.)

Theorem: The general step of the Gaussian elimination

If $a_{k,k}^{(k-1)} \neq 0$, then in Step k we have

$$a_{ij}^{(k)} = a_{ij}^{(k-1)} - \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \cdot a_{kj}^{(k-1)} \quad \begin{array}{l} k = 1, \dots, n-1; \\ i = k+1, \dots, n; \\ j = k+1, \dots, n, n+1. \end{array}$$

Gaussian elimination: 'half way through'

After $n - 1$ steps we have a triangular SLE:

$$A^{(n-1)} = \left[\begin{array}{cccccc|c} a_{1,1}^{(0)} & a_{1,2}^{(0)} & \cdots & a_{1,n-1}^{(0)} & a_{1,n}^{(0)} & a_{1,n+1}^{(0)} \\ 0 & a_{2,2}^{(1)} & \cdots & a_{2,n-1}^{(1)} & a_{2,n}^{(1)} & a_{2,n+1}^{(1)} \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \ddots & a_{n-1,n-1}^{(n-2)} & a_{n-1,n}^{(n-2)} & a_{n-1,n+1}^{(n-2)} \\ 0 & 0 & \dots & 0 & a_{n,n}^{(n-1)} & a_{n,n+1}^{(n-1)} \end{array} \right].$$

Then going backwards: divide the equation with the diagonal element, and then cancel the elements above the diagonal, using analogous row operations.

Finally we arrive at the form $[I|x]$. ($I \in \mathbb{R}^{n \times n}$ is the unit matrix.)

Gaussian elimination: back substitution

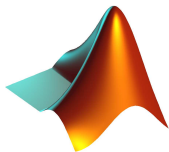
The second (right-to-left) part of the algorithm (i.e. solving a triangular SLE) may alternatively be written using the below formulas. Observe the constant upper indices row-wise.

Theorem: Back substitution formulas

$$x_n = \frac{a_{n,n+1}^{(n-1)}}{a_{n,n}^{(n-1)}},$$
$$x_i = \frac{1}{a_{i,i}^{(i-1)}} \left(a_{i,n+1}^{(i-1)} - \sum_{j=i+1}^n a_{i,j}^{(i-1)} \cdot x_j \right) \quad (i = n-1, \dots, 1).$$

Proof. Write equation i in the triangular SLE and solve for x_i . $\square$

- No heuristics ('oh, I'd like to subtract rather this row from that one'), but **a fixed order of operations!** Suitable for implementation.
- 'Row echelon form': we don't use this term (in this topic). In a sense it is more general, in a sense it is not strict enough for this purpose.
- Compare to Gauss–Jordan elimination. (It is faster this way, $2/3$ times less operations required.)



- 1 Gaussian elimination for 'small' systems ($n \approx 7$)
- 2 The built-in function is of course much faster
- 3 Comparison to the Gauss–Jordan elimination